

Algorithm: Custom Implementation of Support Vector Machine

Dataset: Breast Cancer Dataset (from sklearn.datasets)

1. Business Understanding:

Objective of this study is to develop a Support Vector Machine (SVM) classifier from scratch and compare results with Scikit-learn's own implementation of the SVM classifier, all while trying to classify breast cancer tumors as malignant or benign based on various features input into the model. The main purpose would thus encompass **learning more on SVMs, hyperparameter tuning** (e.g., **the learning rate, lambda regularization, and kernel functions**), and model evaluation with the **accuracy metric**.

The challenges will be ensuring that the numerical stability has been trained well enough, the correct kernel implementation and the efficient optimization of the SVM loss function.

2. Data Understanding

- The Dataset Used: `'load_breast_cancer()'` from Scikit-Learn
- Feature Description:
 - It has about **30 numerical features**, which are the different characteristics that the cell nuclei possess (e.g., **radius, texture, smoothness, compactness**).
- Target variable is binary, `'0'` (**malignant**) or `'1'` (**benign**).
- Initial Observations:
 - The data is ready for classification.
 - It requires feature standardization as the scale of features varies.

3. Data Preparation

- Changes Made: divided into **75% training and 25% testing**
- Feature scaling made with `'StandardScaler'` for normalized data for better convergence of the model.
- Transformed class labels from `'0'` and `'1'` to `'-1'` (malignant) and `'1'` (benign) to match the custom SVM's format.

4. Modelling

4.1 Custom Implementation of SVM

Implementation Details:

- Feature mapping was accomplished with a polynomial transformation kernel (**degree = 3**).
- Optimized using mini-batch Stochastic Gradient Descent (**mini-batch = 10**).

- **Learning Rate = 0.001**, Regularization parameter (**lambda**) = **0.01**.
- Further Number of iterations = **5000**.
- Weights randomized.
- Decision function: **`sign(w . X + b)`**
- Issues with Identified:
- Initial misalignment of shape issues during implementation was subsequently solved using the **transformation of kernel**.
- Improved training stability with appropriate selection of learning rate.

5. Evaluation

Implementation	Accuracy
Custom SVM (Poly)	0.9860
Scikit-learn SVM (RBF)	0.9860

Observations:

Both models are accurate because their outputs are just as close to each other as those from other models, proving the custom implementation is working perfectly. Observed performance change with different kernels or data sets.

Summary:

Successfully created an SVM classifier from scratch using a **polynomial kernel**. **Feature scaling and mini-batch gradient descent** were applied for stable training. The accuracy achieved was identical (**0.9860**) as compared to that of Scikit-learn's SVM with an **RBF kernel**.